Let a further dynamical variable s be defined by

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \xi.$$

The extended Nosé-Hoover system then has a conserved quantity (see Exercise 2):

$$I_{\mathrm{NH}}(\boldsymbol{q},\boldsymbol{p},\xi,s) = H(\boldsymbol{q},\boldsymbol{p}) + \frac{\mu\xi^2}{2} + \mathrm{N_d}\beta^{-1}\ln s. \tag{8.8}$$

This integral can be used to monitor the performance of numerical methods.

8.1.2 Numerical Methods for Extended System Thermostats

We next describe some numerical procedures (based on the concept of splitting introduced in Chaps. 2 and 3) which are appropriate for thermostats based on the incorporation of auxiliary variables.

Nosé-Hoover dynamics is not a Hamiltonian system, despite the presence of the mentioned first integral. It is, however, time-reversible: changing the sign of $\boldsymbol{p}$ and ξ simultaneously in the equations of motion is equivalent to reversing the arrow of time $t \to -t$. For this reason, time-reversible numerical methods are often proposed for Nosé-Hoover. A typical integrator is described below.

To simplify the notation and to use a format that might more closely match what would be implemented in software, we write the algorithm for a single timestep of the method in which positions, momenta and auxiliary variables are updated *in situ*.

Nosé-Hoover Timestep Map

$$\begin{aligned}
\boldsymbol{q} &:= \boldsymbol{q} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p} \\
\boldsymbol{p} &:= e^{-h\xi/2}\boldsymbol{p} - c(h/2,\xi)\nabla U(\boldsymbol{q}) \\
\xi &:= \xi + h\mu^{-1}\left[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T}\right] \\
\boldsymbol{p} &:= e^{-h\xi/2}\boldsymbol{p} - c(h/2,\xi)\nabla U(\boldsymbol{q}) \\
\boldsymbol{q} &:= \boldsymbol{q} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}
\end{aligned}$$

Here we have solved successively, the first, second and third equations, sequentially, and then in reverse order to make a symmetric (and second order) integrator. The auxiliary function involved $c(t,\xi) = \frac{1-e^{-t\xi}}{\xi}$ is easily evaluated from a few terms of its Maclaurin series expansion

$$c(t,\xi) = t - t^2\xi/2 + \ldots$$

It is easy to verify that this method is time-reversible. If we write the timestep map as $(\boldsymbol{q},\boldsymbol{p},\xi) \mapsto \mathscr{G}_h(\boldsymbol{q},\boldsymbol{p},\xi)$, then we may check that

$$\boldsymbol{R} \circ \mathscr{G}_h \circ \boldsymbol{R} = \mathscr{G}_h^{-1}, \qquad \boldsymbol{R}(\boldsymbol{q},\boldsymbol{p},\xi) = (\boldsymbol{q},-\boldsymbol{p},-\xi).$$